

Login to ICE (with VPN)

- <https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard/>

Interactive Apps

Applications

ANSYS Workbench

Jupyter

MATLAB

Ollama + Jupyter (Beta)

RStudio

vLLM + Jupyter

Desktops

Interactive Desktop

IDEs

Coder

Microsoft VS Code

Tutorials

End-to-End-LLM Bootcamp

Makerspace Jupyter Tutorial

Coder

This app will launch [Coder](#) on a compute node. Coder is an IDE based on the open-source components of Microsoft VS Code.

This OOD application will no longer receive updates. Please switch to the [Microsoft VS Code OOD Application](#), which provides the official version of Microsoft VS Code.

Modules

Anaconda 3 - 2023.03

PACE-Apps modules to load before starting VS Code.

- VS Code extensions will only find PACE-apps if modules are specified here!
- To run custom module loads and shell commands, specify "Custom Modules" and fill in the text field below.

Custom Pre-load Commands

Optional. Bash commands to run before starting VS Code. Example: `module load cuda/11.7`

- You can enter multiple lines and arbitrary shell commands.
- Only used when "Custom Pre-load" is selected in the "Modules" dropdown, above

Quality of Service

Default (none)

Node Type

Intel CPU

Cores

6

Number cores (or "CPUs") for session. VS Code is more responsive with ≥ 4 cores.

Connect to Coder

- <https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard/>

Interactive Apps

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Coder (5583880)

1 node | 6 cores | Running

Host: `>_atl1-1-01-005-1-1.pace.gatech.edu`

Delete

Created at: 2026-08-19 23:16:21 EDT

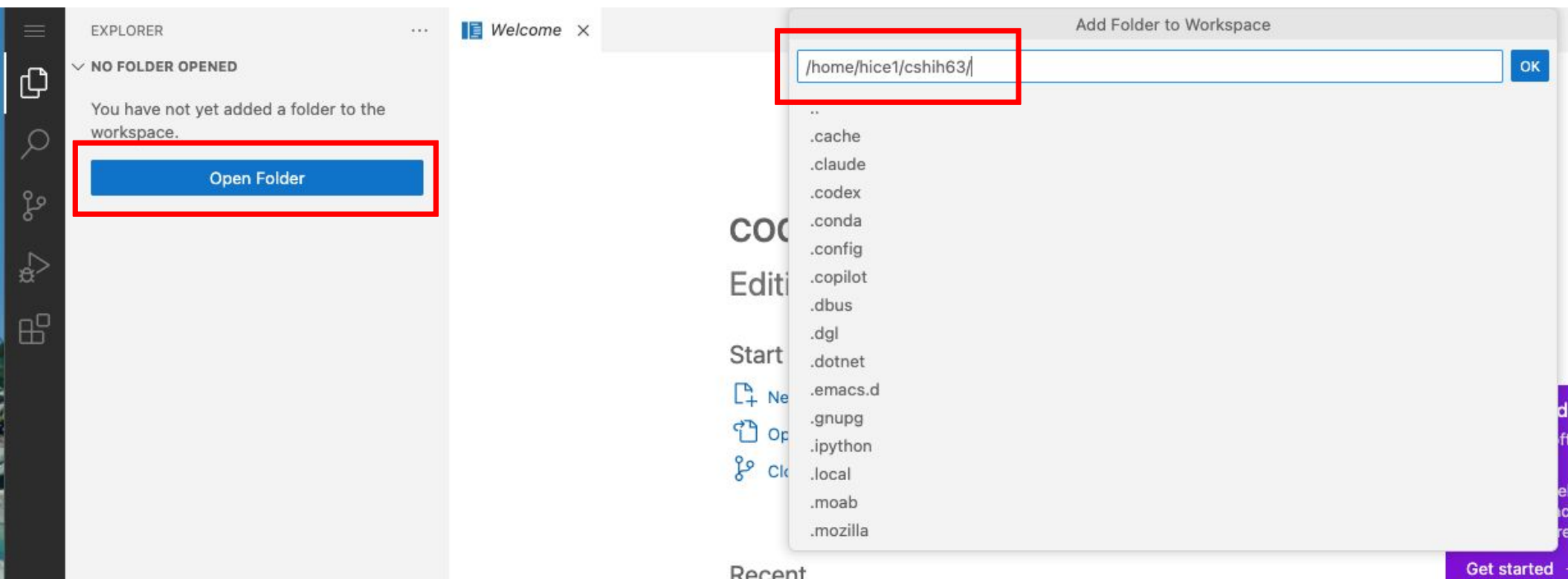
Time Remaining: 3 hours and 58 minutes

Session ID: a8a9baae-10ec-42f4-b736-3cc26f17d07b

Connect

Open Home Directory

- <https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard/>



New Terminals

The image shows the Visual Studio Code (VS Code) interface. On the left, the 'Terminal' menu is open, with 'New Terminal' highlighted. The menu also shows 'Split Terminal', 'Run Task...', 'Run Build Task...', 'Run Active File', 'Run Selected Text', 'Show Running Tasks...', 'Restart Running Task...', 'Terminate Task...', 'Configure Tasks...', and 'Configure Default Build Task...'. The background shows the 'code-server' welcome screen with the text 'code-server Editing evolved', 'Start', 'New File...', 'Open File...', 'Clone Git Repository...', and 'Recent'.

Terminal Menu:

- File
- Edit
- Selection
- View
- Go
- Run
- Terminal**
- Help
- Sign out of code-server
- > .dotnet
- > .emacs.d
- > .gnupg
- > .ipython
- > .local
- > .moab
- > .mozilla
- > .nv
- > .ssh
- > .triton
- > .vnc

New Terminal Sub-menu:

- New Terminal**
- Split Terminal
- Run Task...
- Run Build Task...
- Run Active File
- Run Selected Text
- Show Running Tasks...
- Restart Running Task...
- Terminate Task...
- Configure Tasks...
- Configure Default Build Task...

code-server

Editing evolved

Start

- New File...
- Open File...
- Clone Git Repository...

Recent

- job (Workspace) ~/ondemand/data/sys/dashboard/batch_connect/sys/bc_c...

Check Binaries

- source /storage/ice-shared/cs3220/labs_setup.sh
- With **iverilog** installed, you are ready to go

```
~$ source /storage/ice-shared/cs3220/labs_setup.sh
Lmod has detected the following error: These module(s) or extension(s) exist but cannot be loaded
as requested: "gtkwave"
Try: "module spider gtkwave" to see how to load the module(s).
```

```
~$ iverilog
iverilog: no source files.
```

```
Usage: iverilog [-EiSuvV] [-B base] [-c cmdfile|-f cmdfile]
               [-g1995|-g2001|-g2005|-g2005-sv|-g2009|-g2012] [-g<feature>]
               [-D macro[=defn]] [-I includedir] [-L moduledir]
               [-M [mode=]depfile] [-m module]
               [-N file] [-o filename] [-p flag=value]
               [-s topmodule] [-t target] [-T min|typ|max]
               [-W class] [-y dir] [-Y suf] [-l file] source_file(s)
```

See the man page for details.

```
~$ |
```

Install Extensions - iverilog/gtkwave

EXTENSIONS: MARKETPLACE

gtkwave



Icarus Verilog + GTKWave Runn...
Run Verilog/SystemVerilog testbe...
j054n



dglog
Run Verilog/SystemVerilog testbe...
j054n

Extension: Icarus Verilog + GTKWave Runner for VSCode



Icarus Verilog + GTKWave Runner for VSCode

j054n | 3,384 | ★★★★★

Run Verilog/SystemVerilog testbench with GTKWave support.

Disable

Uninstall

⚙️

This extension is enabled globally.

DETAILS

FEATURE CONTRIBUTIONS

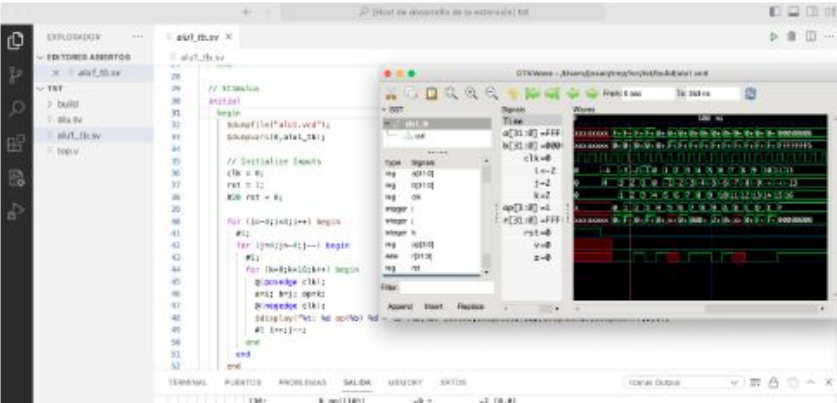
EXTENSION PACK

that **iverilog** and **gtkwave** have been added to your environment PATH variable. If you need help, check out the installation guide for [Windows](#) and [other OSes](#).

Usage

The extension adds in VSCode:

- two buttons in the titlebar of any **Verilog/SystemVerilog** file:
 - Run testbench simulation
 - Stop simulation and clean build files
- a **status** item positioned in the lower-right corner
- **console log output** of commands and running testbench



Categories

Programming Languages

Extension Resources

[Marketplace](#)
[Repository](#)
[License](#)
j054n

More Info

Published1/26/2024,
15:23:36

Last1/26/2024,
released 15:23:36

Identifier dglab.dglog

Install Extensions - SystemVerilog

EXTENSIONS: MARKETPLACE

systemverilog

- Vide - Verilog/SystemVerilog** 620
A Modern SystemVerilog Coding I...
pascal-lab [Install](#)
- Vide - Verilog/System...** 6K ★ 5
A Modern SystemVerilog Coding I...
pascal-lab [Install](#)
- SystemVerilog Pro** 4K ★ 5
Fast SystemVerilog, Verilog, and ...
Taher Anaya [Install](#)
- Verilog-HDL/Syste...** 304K ★ 5
Verilog-HDL/SystemVerilog suppo...
mshr-h [Install](#)
- SystemVerilog Tools** 1K
Verilog/SystemVerilog formatter, ...
JayceVane [Install](#)
- slang-server: Verilog/...** 6K ★ 5
Verilog and SystemVerilog suppor...
Hudson-River-Trading [Install](#)
- SystemVerilog LSP - Synt...** 1K
Simple SystemVerilog language s...
mashurr [Install](#)
- VeriGood - SystemVerilo...** 3K
Verilog/SystemVerilog formatter w...
FabioOliveira [Install](#)
- SystemVerilog and Verilo...** 5K
Beautify SystemVerilog and Verilo...
bmpenuelas [Install](#)
- HDL BlockSeal** 330
Auto-inserts closing keywords (en...
VENOMNBB [Install](#)
- DigitalJS Schematic Vie...** 332
Synthesises Verilog/SystemVerilo...
VENOMNBB [Install](#)

Extension: Verilog-HDL/SystemVerilog x test.v

Verilog-HDL/SystemVerilog v1.29.0

mshr-h | 304,024 | ★★★★★ (1)

Verilog-HDL/SystemVerilog support for VS Code

[Install in ondemand-ice.pace.gatech.edu](#) ⚙️

[DETAILS](#) [FEATURE CONTRIBUTIONS](#) [CHANGELOG](#)

HDL support for VS Code

HDL support for VS Code with syntax highlighting, snippets, classic Ctags-backed editing features, linting, formatting, waveform viewing, inactive preprocessor highlighting, and optional external language servers.

openssf best practices in progress 16% [Ask DeepWiki](#)

```
1 module some_mod(  
2     input wire clk,  
3     input wire rst,  
4     input wire in,  
5     output reg out  
6 );  
7  
8     reg rand_reg;  
9  
10    always@* begin  
11        out = ~in  
12    end  
13  
14 endmodule
```

Categories

- Programming Languages
- Snippets
- Linters
- Formatters
- Visualization

Extension Resources

- [Marketplace](#)
- [Repository](#)
- [License](#)
- [mshr-h](#)

More Info

Published 11/10/2021, 06:04:29

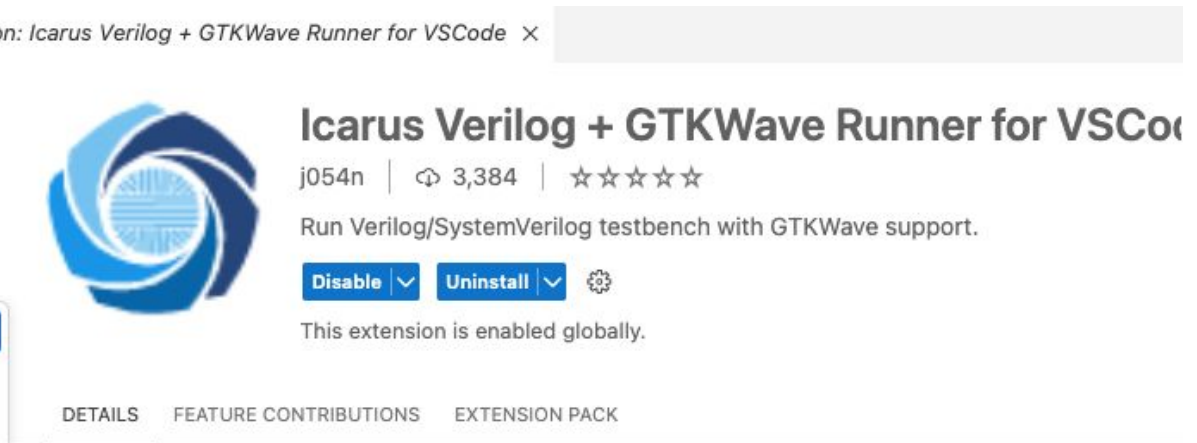
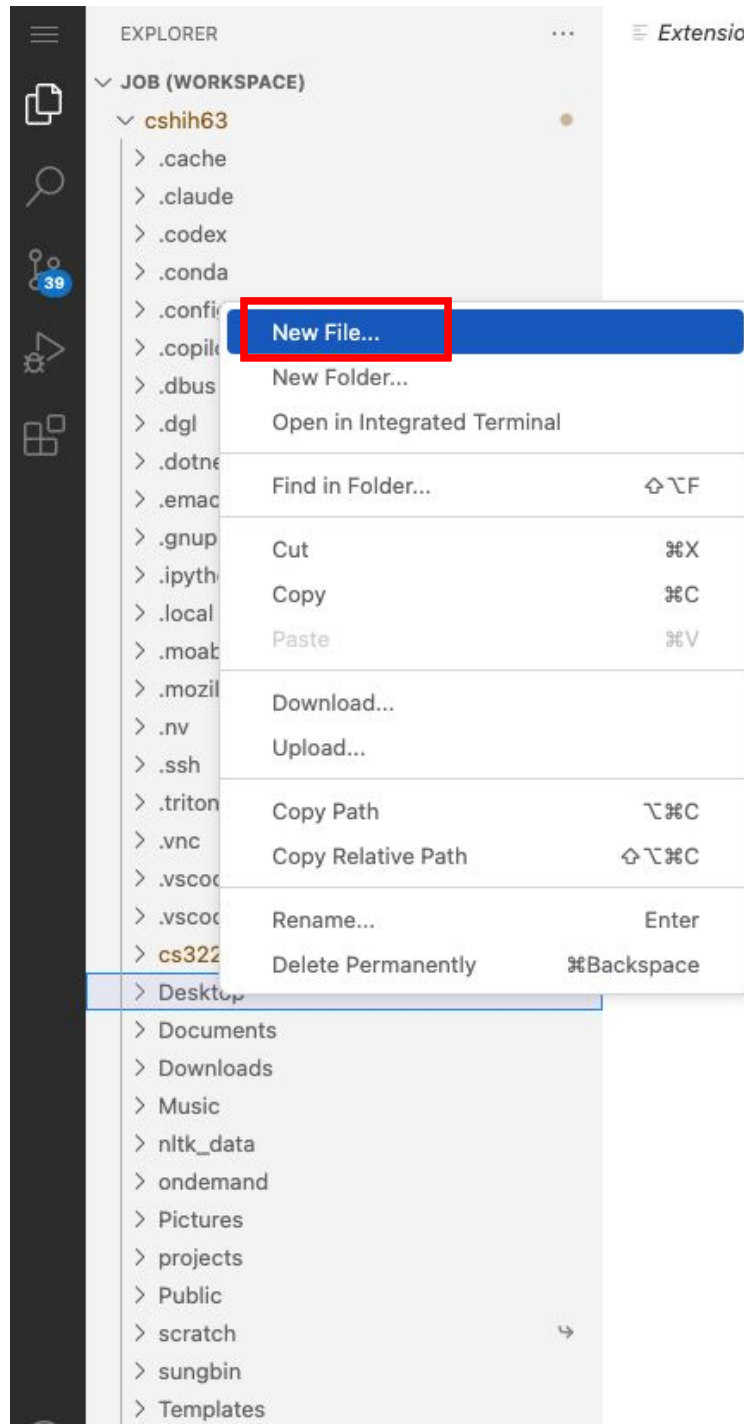
Last released 7/11/2026, 01:34:20

Identifier mshr-h.veriloghdl

Installation

Install the extension from the [VS Code Marketplace](#) or [Open VSX Registry](#)

New File



that **iverilog** and **gtkwave** have been added to your environment PATH variable. If you need help, check out the installation guide for [Windows](#) and [other OSes](#).

Usage

The extension adds in VSCode:

- two buttons in the titlebar of any **Verilog/SystemVerilog** file:
 - Run testbench simulation
 - Stop simulation and clean build files
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Categories

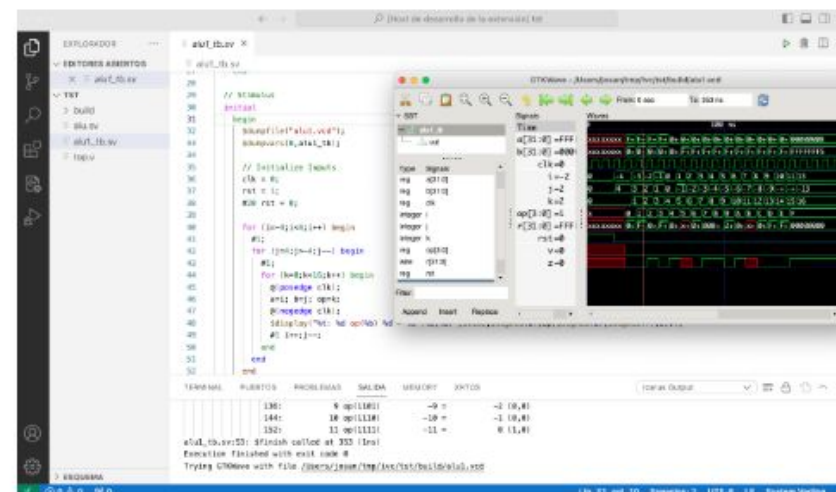
Programming
Languages

Extension Resources

[Marketplace](#)
[Repository](#)
[License](#)
[j054n](#)

More Info

Published 1/26/2024,
15:23:36
Last 1/26/2024,
released 15:23:36
Identifier dglab.dglog



Test Adder

```
1 module testbench;
2     reg [3:0] a;
3     reg [3:0] b;
4     wire [4:0] sum;
5
6     adder dut (
7         .a(a),
8         .b(b),
9         .sum(sum)
10    );
11
12    initial begin
13        $dumpfile("adder.vcd");
14        $dumpvars(0, testbench);
15    end
16
17    integer i;
18    integer errors;
19    initial begin
20        errors = 0;
21        $display("a b sum expected");
22        $display("-----");
23        for (i = 0; i < 8; i = i + 1) begin
24            a = i;
25            b = i + 1;
26            #10;
27            $display("%4d %4d %5d %5d %s",
28                a, b, sum, a + b,
29                (sum == (a + b)) ? "OK" : "FAIL");
30            if (sum != (a + b))
31                errors = errors + 1;
32        end
33        if (errors == 0)
34            $display("PASSED");
35        else
36            $display("FAILED - %0d error(s)", errors);
37        $finish;
38    end
39
40 endmodule
```

testbench.v

```
module adder (
    input [3:0] a,
    input [3:0] b,
    output [4:0] sum
);

    assign sum = a + b;

endmodule
```

test.v

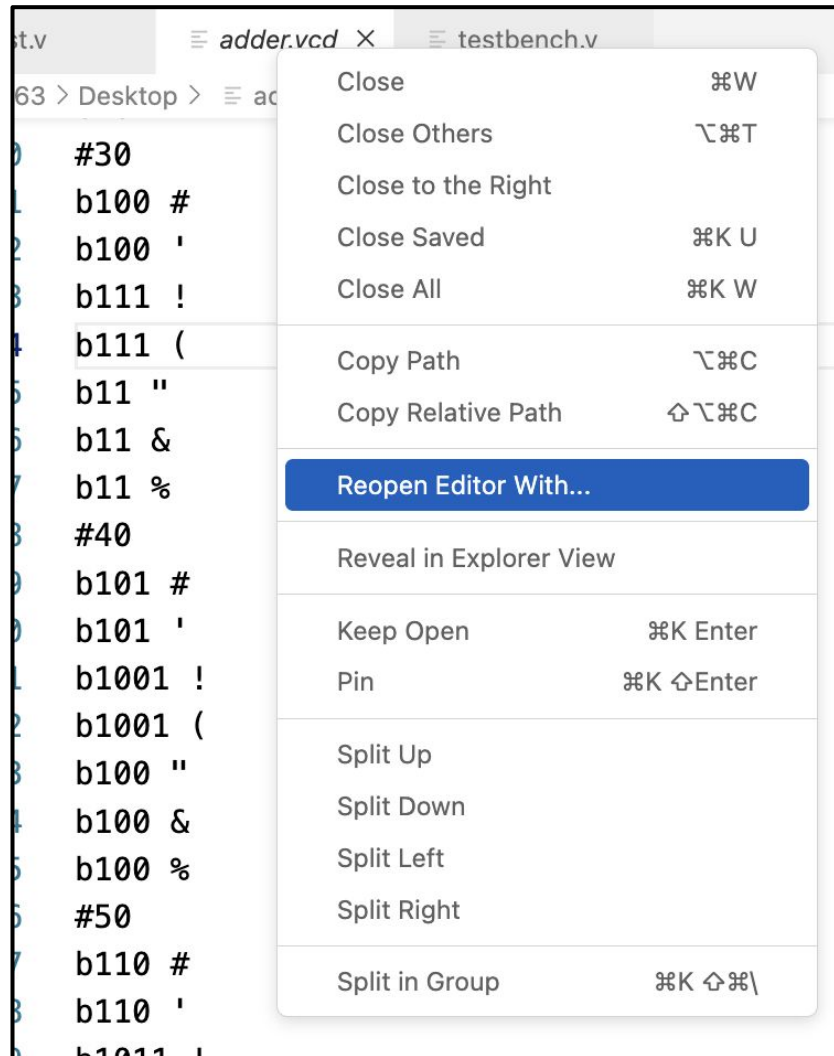
```
~$ cd Desktop/
~/Desktop$ iverilog test.v testbench.v
~/Desktop$ vvp ./a.out
```

a	b	sum	expected	
0	1	1	1	OK
1	2	3	3	OK
2	3	5	5	OK
3	4	7	7	OK
4	5	9	9	OK
5	6	11	11	OK
6	7	13	13	OK
7	8	15	15	OK

PASSED
~/Desktop\$ |

Check waveform

- ❏ Right click vcd file -> Reopen Editor ...
-> Signal Viewer



Check waveform

❏ On the right panel, Add New View

The screenshot displays a digital logic simulation interface. On the left, a 'Value Change Dump' panel shows a hierarchy of testbench components: testbench, dut, sum, a, b, errors, and i. The main area shows a waveform with a time axis from 0 to 50ns. The waveform data is as follows:

Signal	0	10s	20s	30s	40s	50s
x01	x0	x1	x2	x3	x4	x5
x03	x0	x1	x2	x3	x4	x5
x05	x0	x1	x2	x3	x4	x5
x07	x0	x1	x2	x3	x4	x5
x09	x0	x1	x2	x3	x4	x5
x00000000	x00000000	x00000001	x00000002	x00000003	x00000004	x00000005

On the right, a 'View' panel is open, showing the 'add hierarchy' view. The panel includes a 'Name' field with 'add hierarchy', a 'Description' field with 'Enable', and a 'Star' icon. The 'Axis type' is set to 'Linear'. The 'Domain base' is set to 'Use signal domain base'. The 'Mode' is set to 'Use signal domain base'. The 'Domain class' is set to 'Time'. A red box highlights the 'Create a hierarchy of all signals and scopes' button, and a red arrow points to it from the 'Add a flat list of all signals' button.

Or download the vcd file and use gtkwave

Download Files

